



Zhengyang Wang (Alvin)

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EDUCATION

- **Beijing Normal-Hong Kong Baptist University (BNBU)** Zhuhai, China
B.Sc. (Honours) in Computer Science and Technology Sep 2023 – Jun 2027
 - **GPA:** 3.24/4.00 | **Medium of instruction:** English
 - **Core Courses:** **Math & Analytics:** Probability and Statistics (A); Data Analytics for Business (A-)
AI & Computer Vision: 3D Computer Vision (A-); Machine Learning (B+);
Neural Networks and Deep Learning (B+)
CS Foundations: Data Structures and Algorithms (B+);
Database Management Systems (B+); Software Engineering (B+)

RESEARCH EXPERIENCE

- **Research and Development of Third-Generation Intelligent Arrow Selection Equipment and Systems ** Zhuhai, China
Research Assistant / Digital Sports Research Centre, BNBU May 2025 – May 2026
(Supervisors: Huaxiong Huang, DKU; Lidong Fang, SUFE)
 - Organised and coordinated archery shooting trials; curated and manually validated a multimodal dataset linking arrow measurements, target images, impact coordinates and ring scores to assess shot dispersion and stability across rounds for data-driven arrow selection
 - Extracted launch velocity and angle from high-speed video to analyse how shot kinematics relate to impact patterns
 - Led the fine-tuning and integration of SAM 3.1 and YOLO11s for archery target segmentation and arrow impact localisation, achieving 92.58% precision, 88.51% recall, and 90.50% F1 via offline temporal fusion
 - **Manuscript:** Co-author of *Arrow Selection for Elite Series Arrows: Data Driven Archery Arrow Selection*, under peer review at *Sports Engineering*
- **MetroHeight: Mobile RGB-D Cargo Height Estimation** Apr 2026 – Present
(First author / Supervisor: Xiubin Fan, BNBU)
 - Built an iOS capture app and FastAPI backend for cargo-height inspection of delivery trolleys in metro stations
 - Aligned camera intrinsics to depth-map resolution, filtered invalid and low-confidence depth samples, and used DBSCAN clustering to isolate cargo geometry before height estimation
 - Implemented SAM 3 segmentation and gravity-aligned 3D reconstruction, combining local RANSAC floor fitting with robust cargo-top estimation to measure floor-to-cargo height while excluding trolley structures
 - Evaluated 194 capture records across four trolley categories at Hong Kong MTR's Wong Tai Sin Station against laser references, achieving 2.84 cm MAE and 4.27 cm RMSE from estimates averaged over five runs per capture
 - **Manuscript:** First author of *MetroHeight: Smartphone RGB-D Measurement of Cargo Height in Metro Stations*; manuscript in preparation for submission to *ACM SenSys 2027*
- **Spatiotemporal Flood Forecasting and Hydrological Modelling** Sep 2025 – Sep 2026
Research under the Guangdong Major Project of Basic and Applied Basic Research:
High-Resolution Simulation and Prediction of the Land Surface Environment (Project No. 2021B0301030007)
(Supervisors: Huaxiong Huang, DKU; Zijian Wang, BNBU)
 - Developed a flow-direction-aware river-discharge forecasting method combining hydrological time series with spatial river-network graphs to represent downstream hydrological propagation
 - Integrated ERA5-Land reanalysis, hydrometric observations from 75 gauges across 15 rivers, and static geospatial rasters into river-specific 29-channel grids, with water levels forward-filled for up to seven days
 - Implemented event-aware raster-Transformer training with river-balanced regression and high-flow underprediction penalties, and assessed three-seed ensembles using chronological splits and validation-based checkpoint selection
 - **Conference:** Second author of *A Spatiotemporal Hybrid Framework for Regional Flood Forecasting*; accepted for poster presentation at WRE 2026, Seoul, Dec 2026

- **Visual Metaphor and Multimodal Reasoning** Dec 2025 – May 2026
(Project Mentor: Gailun Zeng, Tencent)
 - Led MetaVision-DB data construction, covering 800 images, four visual–semantic relations and three communication games, with concept–image alignment, schema validation and manual quality review
 - Co-developed MACH, a dual-block VLM agent framework combining contrastive visual-semantic reasoning and strategy-conditioned metaphor generation with structured memory, theory-derived knowledge and simulated teammate and opponent responses
 - Evaluated MACH in Visual Undercover, where three MACH civilian agents achieved a 97.2% win rate against two Raw-VLM undercover agents using Gemini 3.1 Pro
 - **Manuscript:** Third author of *Orchestrating Secrets: Visual Metaphor Generation and Reasoning in Multi-Agent Communication Games*; in revision for ARR resubmission
- **MapAgent: Indoor Localisation and Spatial Reasoning** Jul 2026 – Present
(Supervisor: Xiubin Fan, BNBU)
 - Developed a Qwen3-based localisation agent to explore LLM-based spatial reasoning over BLE observations, position-history cues and map constraints, with independent deterministic verification and fallback
 - Evaluated a Radio-Spatial Graph pipeline on 41 trajectories across six sites (2,125 BLE windows); under leave-one-site-out evaluation, walkable-space projection reduced trajectory-averaged error from 3.67 m to 3.49 m
- **Cardiac Segmentation and Spatiotemporal Modelling** Mar 2026 – Aug 2026
 - Developed ST-Net for $2D + t$ cine MRI segmentation with a bottleneck Temporal Attention Gate and second-order cyclic regularisation of area and centroid trajectories under ED/ES-only supervision
 - Benchmarked seven model and loss configurations on 50 ACDC cine MRI cases using five-fold patient-level cross-validation
 - **Manuscript:** Third author of *ST-Net: Temporal Recalibration for Cardiac Cine Segmentation under ED/ES-only Labels*; submitted to AISTATS 2027
- **Hydrological Constraints on Post-Fire Vegetation Recovery** Oct 2025 – Sep 2026
 - Analysed modelled soil-water saturation and MODIS NDVI across equal six-year pre- and post-fire periods, using the 2019–2020 Gaspers Mountain megafire as a natural experiment to assess hydrological constraints on vegetation recovery
 - Quantified an increase in positive soil-water–NDVI trend concordance from 37.8% before fire to 95.3% during 2020–2025 recovery, while finding no positive coupling in year-to-year changes
 - **Conference:** Second author of *Long-term hydrological support constrains vegetation recovery after major disturbance*; accepted for in-person poster presentation at BES Annual Meeting 2026

INTERNSHIPS & OPEN SOURCE

- **Tencent Rhino-Bird Open Source Talent Program** Aug 2026
 - Delivered KuiklyLoadingKit , a Kotlin Multiplatform loading component for Tencent KuiklyUI with a declarative DSL and Android/iOS demos
- **Bank of Guangzhou, Hengqin Branch** Zhuhai, China
Corporate Finance Department Intern Jan 2025 – Feb 2025
 - Organised corporate lending and credit records in Excel, supporting document review and information management
- **Beijing Normal-Hong Kong Baptist University (BNBU)** Zhuhai, China
Finance Office Intern — Work-study Programme Sep 2024 – Dec 2024
 - Supported expense reimbursement, voucher organisation and document filing

HONOURS & AWARDS

- **Dean's List** (five semesters), BNBU – 2023–2026
- **Excellence Award**, ICBC Cup University FinTech Innovation Competition – 2023
- **Guangdong Provincial Excellence Award**, China Undergraduate Mathematical Contest in Modeling (CUMCM) – Oct 2025
- **Bronze Award of Servant Leadership**, Student Leadership Path Programme, BNBU – May 2026
- **NVIDIA Certificate of Competency**, Getting Started with Deep Learning – Jul 2026

LEADERSHIP & ACTIVITIES

- **Student Union Staff**, Student Rights & Welfare Department, BNBU – 2023–2024
- **Organisation Officer**, Communist Youth League Class Branch, BNBU – 2023–2024
- **Co-founder**, BNBU Volleyball Club

TECHNICAL SKILLS & LANGUAGES

Skills: Python, SQL, Swift, Kotlin, FastAPI, PyTorch, Hugging Face, ModelScope, OpenCV, Git, LaTeX

Languages: Mandarin (Native); English (IELTS Academic Overall 6.5; CET-4 and CET-6)